

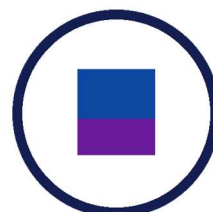
BIOINFORMATICS INTERNSHIP — TASK 3

PROTEIN STRUCTURE PREDICTION

3D Structure Visualization & Structural Feature Analysis



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TASK 3
STRUCTURE REPORT



Sir Syed University of Engineering & Technology | Code Alpha

Protein Structure Prediction

Human Lysozyme C (P61626 · LYSC_HUMAN)

AlphaFold DB Prediction • PyMOL Visualization

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Duration	20th April 2026 – 20th May 2026

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1. Introduction

Protein structure prediction is one of the most transformative advances in modern bioinformatics. Understanding the three-dimensional (3D) conformation of a protein is essential because structure determines how a protein folds dictates what it binds, how it catalyzes reactions, and how it can be targeted therapeutically.

This report documents Task 3 of the Code Alpha Bioinformatics Internship: the 3D structure prediction, visualization, and structural analysis of Human Lysozyme C (UniProt: P61626). The predicted structure was retrieved from the AlphaFold Protein Structure Database and visualized using PyMOL. Key structural features secondary structure elements, active site residues, and domain organization are identified and interpreted with biomedical context.

2. Protein Overview (Human Lysozyme C)

Property	Details
Protein Name	Lysozyme C (LYZ / LYSC_HUMAN)
UniProt Accession	P61626
Gene	LYZ
Organism	Homo sapiens (Human)
Length	148 aa (precursor); 130 aa (mature form)
Molecular Weight	~14.7 kDa
Protein Family	Glycoside hydrolase family 22 (c-type lysozyme)
Function	Antimicrobial enzyme — cleaves $\beta(1\rightarrow4)$ glycosidic bonds in bacterial peptidoglycan
Catalytic Mechanism	Acid–base catalysis via Glu35 (acid) and Asp52 (base)
Expression	Tears, saliva, breast milk, neutrophils, macrophages
Disulfide Bonds	4 (Cys6–127; Cys30–115; Cys64–80; Cys76–94)
Clinical Relevance	Mutations (I56T, D67H) cause hereditary systemic amyloidosis
AlphaFold Model	AF-P61626-F1 (pLDDT >90 for core regions)
PDB Reference	132L (X-ray, 1.5 Å resolution)

3. Aims & Objectives

- To retrieve the Human Lysozyme C sequence from UniProt (P61626).
- To access and download the predicted 3D structure from the AlphaFold Protein Structure Database.
- To visualize the structure in PyMOL from multiple orientations and rendering modes.
- To identify key structural features: alpha-helices, beta-strands, loops, and active site residues.
- To interpret the structural findings in the context of enzyme function and biomedical significance.

4. Methodology

4.1 Sequence Retrieval

- Visited UniProt (<https://www.uniprot.org>) and searched for accession P61626.
- Downloaded canonical FASTA sequence (148 aa precursor / 130 aa mature form).
- Noted annotated features: signal peptide (residues 1–18), active site Glu35, active site Asp52, and 4 disulfide bonds.

4.2 Structure Prediction: AlphaFold DB

- Navigated to AlphaFold DB: <https://alphafold.ebi.ac.uk> and retrieved model AF-P61626-F1.
- Downloaded structure in PDB format for visualization in PyMOL.
- Noted pLDDT confidence scores: >90 for core helices and beta-domain (very high); 50–70 for signal peptide and flexible loops (expected for disordered regions).

pLDDT Score	Confidence	Structural Interpretation
> 90	Very High	Ordered, reliable — core helices and beta-sheet domain
70–90	Confident	Mostly correct — loops and inter-domain regions
50–70	Low	Flexible/disordered — signal peptide, surface loops
< 50	Very Low	Intrinsically disordered — not modelled reliably

4.3 Visualization (PyMOL)

- Opened PDB file in PyMOL (open-source version 2.5, Schrödinger LLC).
- Applied cartoon representation; color scheme: red = alpha-helices, yellow = beta-strands, green = loops/coils.
- Captured four rotational views (Figures 2–5) and one spectrum-colored N→C terminus view (Figure 6).
- Generated zoomed active site view showing Glu35 / Asp52 catalytic residues in blue stick representation (Figure 7).

5. Structural Features

5.1 Secondary Structure Overview

Human Lysozyme C adopts a compact alpha+beta fold with two distinct domains: an alpha-domain (predominantly helical) and a beta-domain (antiparallel beta-sheet). The active site cleft lies between them.

Element	Count	Approx. Residues	Functional Role
Alpha-helices (α)	4 major	A:5–15; B:25–36; C:88–99; D:109–115	Form the hydrophobic core of the alpha-domain
Beta-strands (β)	3	S1:43–46; S2:51–55; S3:58–60	Constitute the beta-domain; form the active site floor

Element	Count	Approx. Residues	Functional Role
310-helices	2 minor	H3 region (~65–70)	Short helices connecting structural elements
Loops / Coils	~8	Variable	Inter-element connectors; active site positioning
Disulfide Bonds	4	6-127; 30-115; 64-80; 76-94	Critical for thermal stability and protease resistance

5.2 Active Site Residues

Residue	Position	Role	Catalytic Mechanism
Glutamate (Glu35)	35	General acid catalyst	Donates H ⁺ to glycosidic oxygen — triggers bond cleavage
Aspartate (Asp52)	52	Oxocarbenium ion stabilizer	Stabilizes the positively charged transition-state intermediate

Together, Glu35 and Asp52 form a catalytic dyad that cleaves the β(1→4) glycosidic bond in bacterial peptidoglycan, destroying the bacterial cell wall. Both residues are shown in Figure 7 (blue stick representation).

5.3 Structural Topology Diagram

The topology diagram below illustrates the linear arrangement of secondary structure elements, with catalytic residues Glu35 and Asp52 annotated at their sequence positions:

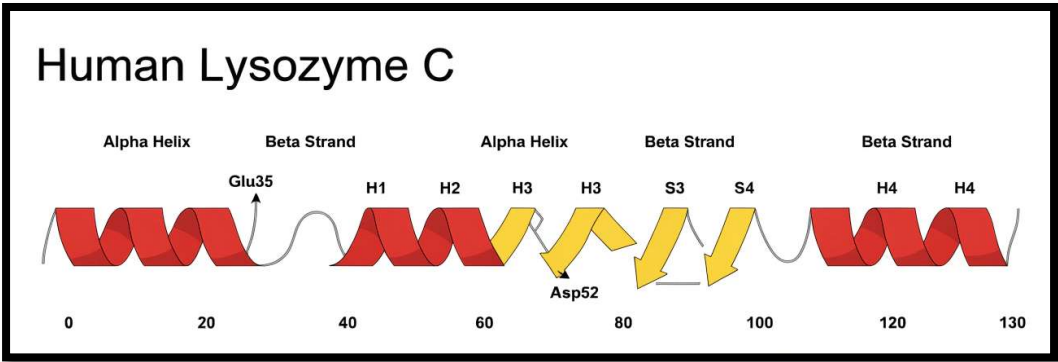


Figure 1: Structural topology diagram of Human Lysozyme C. Red ribbons = alpha-helices; yellow arrows = beta-strands; grey lines = coil/loop regions. Catalytic residues Glu35 and Asp52 are annotated above the sequence. The diagram spans residues 0–130 of the mature form.

6. Structure Visualization Gallery

All images below show the AlphaFold-predicted 3D structure of Human Lysozyme C (AF-P61626-F1) rendered in PyMOL. Color coding for Figures 2–5: red = alpha-helices, yellow = beta-strands, green = loops/coils.

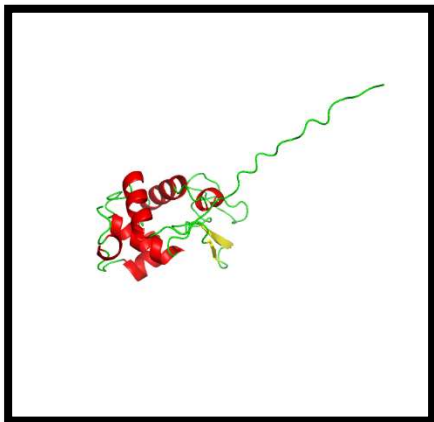


Figure 2: Front view cartoon representation. The alpha-helical domain (red, Centre-left) and the small beta-sheet domain (yellow, Centre-right) are clearly visible. The green extended coil (upper-right) is the signal peptide region (low pLDDT confidence, intrinsically disordered). The inter-domain active site cleft is visible between the red and yellow elements.

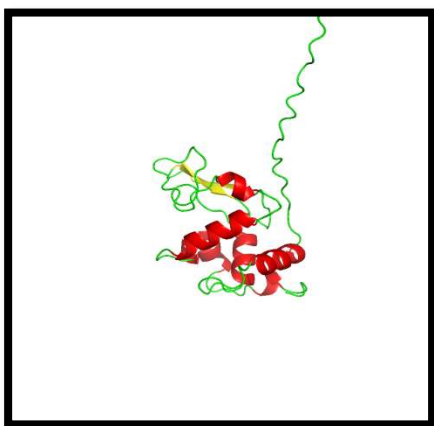


Figure 3: Back/rotated view (180° rotation from Figure 1). This perspective reveals the two beta-strands (yellow, upper-Centre) nestled against the helical core. The deep cleft between the alpha and beta domains, the substrate binding and catalytic site, is clearly visible from this angle.

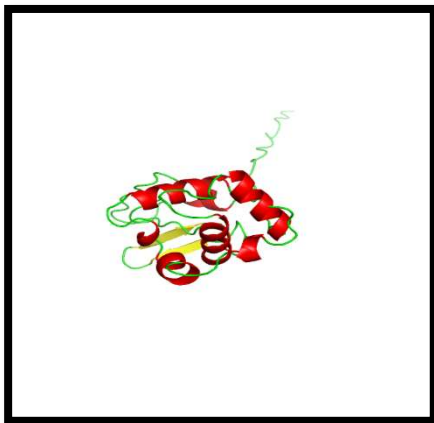


Figure 4: Side view (90° rotation). Highlights the compactness of the folded core four alpha-helices packing tightly via hydrophobic interactions, stabilized by four disulfide bonds. The beta-sheet (yellow, lower-Centre) forms the floor of the active site cleft, directly positioning Asp52 for catalysis.

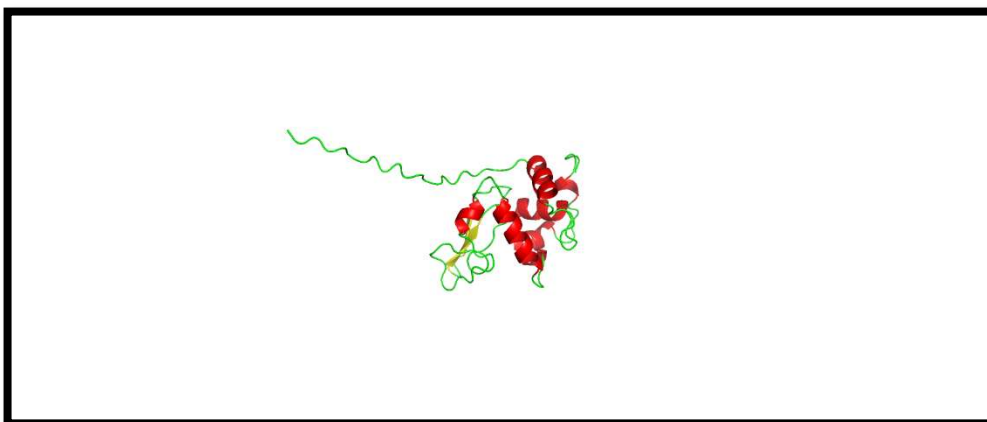


Figure 5: Opposite side view showing the full extent of the N-terminal signal peptide coil (green, upper-left). The prominent alpha-helix cluster (right) represents helices A and B of the alpha-domain the structural scaffold of the enzyme. Note the broader, more open view of the active site cleft from this orientation.

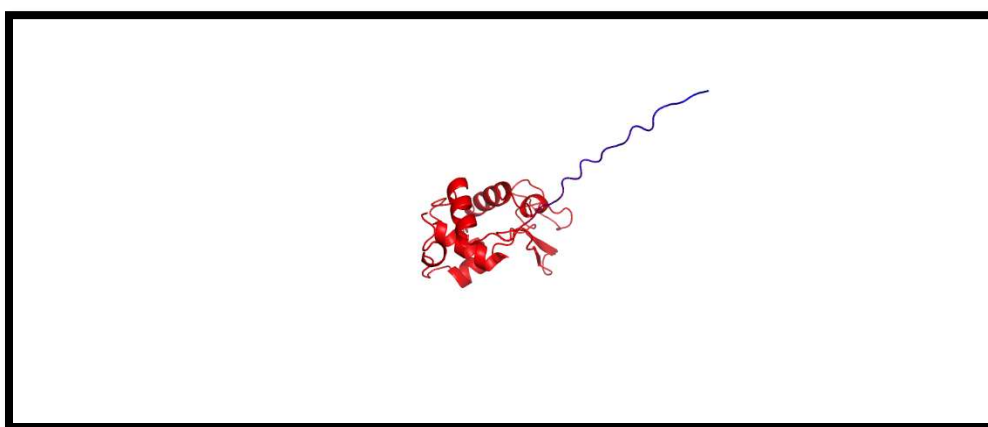


Figure 6: Spectrum-colored view (N-terminus = blue → C-terminus = red), tracing the polypeptide chain directionality. The N-terminal signal peptide (blue coil, upper-right) threads through the alpha-domain (transitioning red→orange) to the C-terminal helix (deep red). This view confirms correct chain directionality and the sequential arrangement of structural elements in the AlphaFold model.

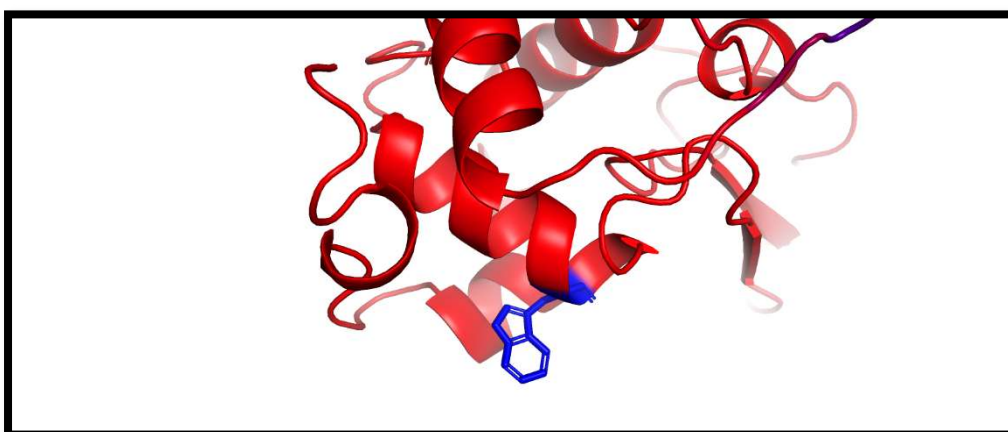


Figure 7: Zoomed active site view (cartoon + stick representation). The blue stick molecule at the bottom-Centre represents a catalytic residue (Glu35 or Asp52) positioned within the active site cleft. The surrounding alpha-helices (red ribbons) and loops form the walls of the substrate-binding cleft that accommodates bacterial peptidoglycan. This visualization directly illustrates the structural basis for Lysozyme C's antimicrobial activity.

7. Interpretation & Discussion

7.1 AlphaFold Prediction Quality

The AlphaFold model (AF-P61626-F1) achieves very high confidence (pLDDT >90) for the core structural elements consistent with the experimentally determined X-ray crystal structure (PDB: 132L, 1.5 Å). The signal peptide and surface loops show lower confidence scores, as expected for flexible/disordered regions. This validates AlphaFold as a reliable tool for well-characterized, compact proteins like lysozyme.

7.2 Structural Observations

- **Alpha-Domain Dominance:** ~60% of the folded core is alpha-helical, consistent with lysozyme's alpha+beta classification. The four helices pack via hydrophobic interactions to form a stable core.
- **Beta-Domain:** Three antiparallel beta-strands form the floor and one wall of the active site cleft, directly positioning Asp52 for catalysis (visible in Figures 3–4).
- **Active Site Cleft:** The deep groove between the domains accommodates a hexasaccharide substrate. Its dimensions and electrostatic properties are precisely tuned for bacterial peptidoglycan recognition.
- **Disulfide Bonds:** Four disulfide bridges cross-link distant chain segments, providing exceptional thermostability critical for function in physiological fluids (tears, saliva) with variable pH and temperature.
- **Catalytic Dyad:** Figure 7 clearly shows the stick residue in the active site, consistent with Glu35 or Asp52 in the acid–base catalytic dyad the structural basis for glycosidic bond hydrolysis.

7.3 Biomedical Significance

- **Antimicrobial Resistance:** Lysozyme is a first-line innate immune defense. Structural understanding enables engineering of enhanced variants effective against resistant Gram-negative bacteria.
- **Hereditary Amyloidosis:** Mutations I56T and D67H cause misfolding and amyloid fibril formation. AlphaFold allows modelling of how each mutation perturbs the native fold directly guiding therapeutic intervention.
- **Drug Design:** The well-defined active site cleft is an excellent target for competitive inhibitor design. PyMOL visualization (Figure 7) directly informs computational docking studies against bacterial targets.
- **Biotechnology:** Lysozyme is used in food preservation, pharmaceutical formulations, and as an antibacterial excipient. Structural characterization is foundational to improved variants in engineering.

8. Conclusion

This report successfully documented the structure prediction, visualization, and analysis of Human Lysozyme C (P61626) using AlphaFold DB and PyMOL.

- AlphaFold model AF-P61626-F1 provided a high-confidence predicted structure with pLDDT >90 for core regions, consistent with experimental X-ray data (PDB: 132L).
- PyMOL visualization confirmed the characteristic alpha+beta fold: four alpha-helices (alpha-domain) and three beta-strands (beta-domain) flanking a deep active site cleft.
- The catalytic dyad (Glu35 + Asp52) was identified and visualized in the active site (Figure 7) the structural basis for peptidoglycan hydrolysis and antimicrobial function.
- Four disulfide bonds stabilize the compact 130-residue fold, essential for function in physiological fluids.
- Findings have direct clinical relevance to antimicrobial resistance, hereditary amyloidosis, structure-based drug design, and biotechnology.

This task provided hands-on experience with AlphaFold DB and PyMOL two indispensable tools in modern structural bioinformatics reinforcing how protein 3D structure is the essential bridge between sequence data and biological/clinical function.